Flap Actuator – Maintenance Practices

1. Removal/Installation

Note This procedure requires a minimum of two people.

A. Remove Wing Flap Actuator

Refer to Figure 1 and Figure 2 for details.

1) Remove forward cabin ceiling panels.

2) Disengage HYDRAULIC OIL PUMP circuit breaker and install locking clip. Discharge hydraulic pressure from accumulators by operating brake pedals.

3) Move flap selector lever to flaps up position, and pump flaps up, using hand pump.

4) Disconnect flap/elevator interconnect ball screw jack from right inboard bellcrank.

Note Tape cables to flap interconnect ball screw jack cable drum to prevent cable unseating from pulley grooves, and inadvertent movement of ball screw jack.

5) Move flap selector lever to 20° down position. Disengage flap/interconnect ball screw jack from bellcrank by pumping flaps down, using hand pump.

6) Support flaps and remove bolts securing left and right push-pull rods to inboard bellcranks.

7) Lift flaps manually to withdraw push-pull rods from inboard bellcranks. Support flaps at this position.

8) Move flap selector lever to up position.

9) Remove flap actuator forward drip tray.

10) Identify up and down selector cables, and remove selector cable terminals from selector valve pulley.

11) Disconnect hydraulic lines at selector valve. If fitted, remove ski hydraulic lines routed through A-frame. Cap lines (refer to 29-00-00, Hydraulic Power – General Data).

12) Release hydraulic line clips from flap actuator support channel.

13) Remove four bolts and washers securing A-frame to airframe structure.

14) Support flap actuator and A-frame assembly, and remove six bolts and washers securing flap actuator support channel to airframe structure.

15) Remove flap actuator and A-frame assembly from aircraft.

16) Slacken selector valve follow-up cable at A-frame pulley adjusting nut, and remove terminals from selector valve pulley. Release cable from flap actuator pulley and flap actuator support channel.

17) Remove follow-up cable lever, links from flap actuator piston rod end fitting.

18) Remove flap actuator support channel, cushion and clamp from flap actuator.

19) Remove A-frame assembly and aft drip tray from flap actuator.

20) Remove hydraulic lines between flap selector valve and actuator. Cap ports.

B. Install Wing Flap Actuator

Refer to Figure 1 and Figure 2 for details.

1) Position A-frame and aft drip tray on flap actuator. Coat attachment bolts with corrosion protective compound, Shell CG929 (refer to 20-30-01, Protective Coatings Hard and Soft Film Corrosion Preventive Compound – General Data).

2) Secure A-frame and aft drip tray to flap actuator, using two bolts, washers and nuts.

   Note Bolts to be installed with heads up.

3) Insert links in flap actuator piston rod end fitting, with two teflon washers in position.

   Note Install teflon washer between lower face of each link and piston rod end fitting.

4) Secure links to piston rod end fitting as follows:

   a) Position follow-up cable lever on bottom face of piston rod end fitting, with cable and retainer aligned with actuator pulley, and lever flange facing down.

   b) Coat attachment bolts with corrosion protective compound, Shell CG929. Secure indicator post, links and follow-up cable lever to piston rod end fitting using two bolts, washers, nuts and cotter pins.

      Note Bolts to be installed with heads up.

5) Install cushion and support channel on actuator. Secure with clamp but do not fully tighten at this stage. On aircraft incorporating Mod 6/1748, loosen four bolts joining support channels. Install cushion and support channels on actuator and secure with clamp. Ensure that bottom edges of support channels are parallel.

6) Ensure selector lever and selector valve pulleys are in flaps-up position, and flap actuator piston is fully retracted (refer to Figure 2).

7) Route upper portion of follow-up cable under flap actuator pulley and over selector valve pulley. Connect cable terminal to selector valve follow-up pulley.

8) Route remaining portion of follow-up cable through hole in flap actuator support channel and under flap selector valve follow-up pulley.

9) Connect follow-up cable terminal to flap selector follow-up pulley.

10) Remove caps from flap actuator ports, position suitable container to receive spillage and extend flap actuator piston approximately 2.0 inches. Reinstall caps.

11) Finally, position flap actuator and A-frame assembly in aircraft. Secure support channel with six bolts and washers, and A-frame with four bolts, washers and nuts.

12) Tighten clamps securing actuator to support channel. Clip hydraulic lines to flap actuator support channel. On aircraft incorporating Mod 6/1748, tighten four bolts joining support channels, allowing actuator to retain its free position as determined by anchor bolts at rear frame.

13) Connect inboard bellcranks to push-pull rods as follows:

   a) Remove caps from flap actuator ports, position container to receive spillage, and move flap actuator piston until bellcranks engage with push-pull rods. Replace caps.

   b) Secure bellcranks to push-pull rods with two bolts, four washers, two nuts and two cotter pins.

      Note Bolts to be installed heads down with one washer positioned under each bolt head.

14) Ensure selector valve selector pulley is in flaps-up position (refer to Figure 2). Connect flap selector cable terminals to selector pulley and lockwire in position.

    Note Flap-down selector cable is routed over selector pulley.

15) Tension flap selector and follow-up cables (refer to 27-00-00, Flight Controls – General Data). Ensure position of selector valve pulleys remain unchanged. Lock turnbuckles.

    Note Adjust tension of follow-up cables at A-frame pulley adjustment nut.

16) Remove caps, connect hydraulic lines to flap selector valve, and install hydraulic lines between actuator and selector valve.

17) Using hand pump, pump flaps up and guide right bellcrank into fork end of flap/interconnect ball screw jack.

18) Install one nylon washer between upper and lower faces of bellcrank and flap/interconnect ball screw jack fork end.

19) Coat bellcrank and flap/interconnect ball screw jack attachment bolt with corrosion, protective compound, Shell CG929. Install bolt, and secure with washer, nut and cotter pin.

    Note Bolt to be installed with head up.

20) Remove tape from flap/interconnect ball screw jack cable drum.

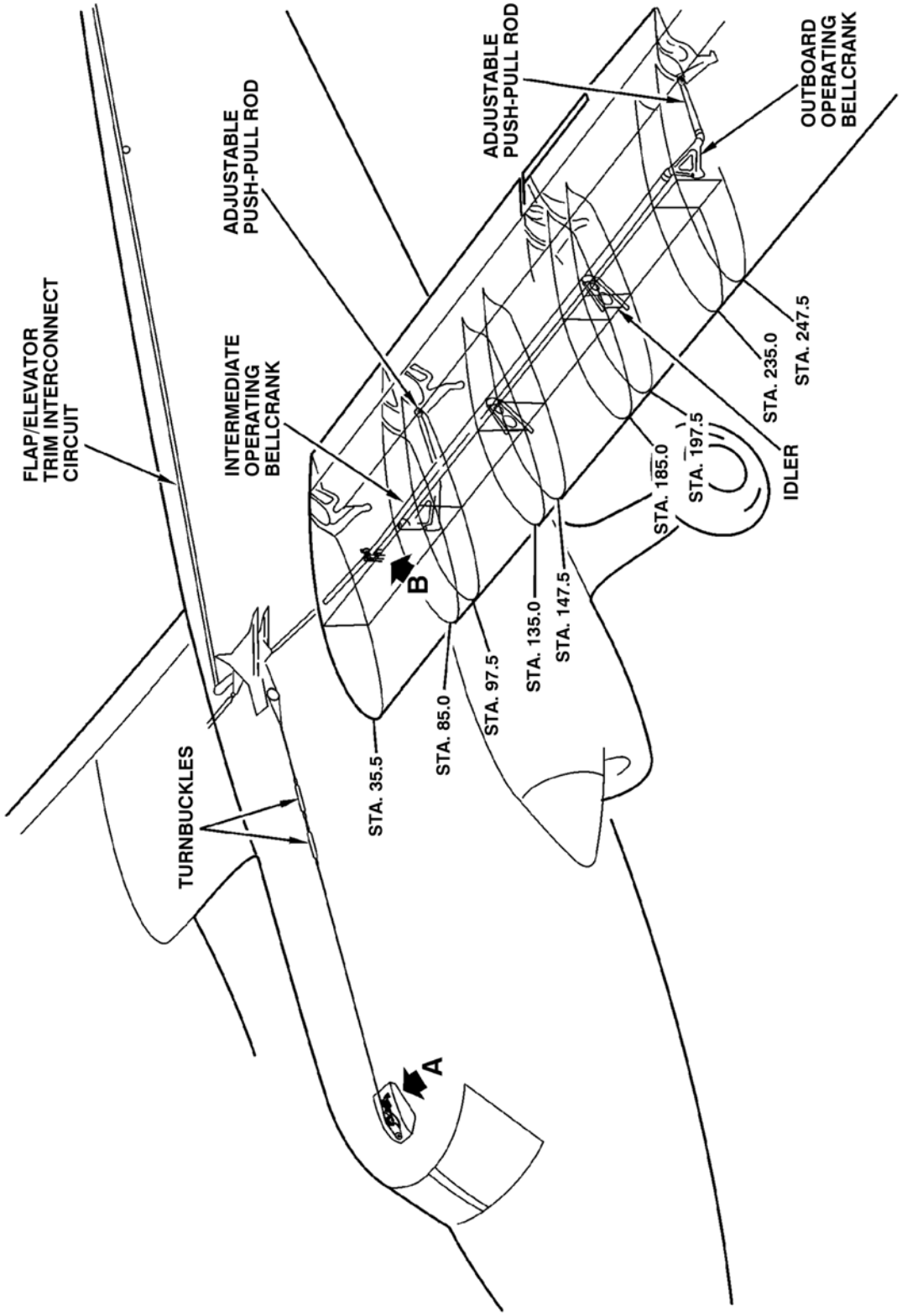
21) Rig flaps (refer to 27-50-00, Wing Flaps System – General Data).

22) Bleed wing flap system (refer to 27-50-00, Wing Flaps System – General Data).

23) Do an operational test on wing flap system (refer to 27-50-00, Wing Flaps System – General Data).

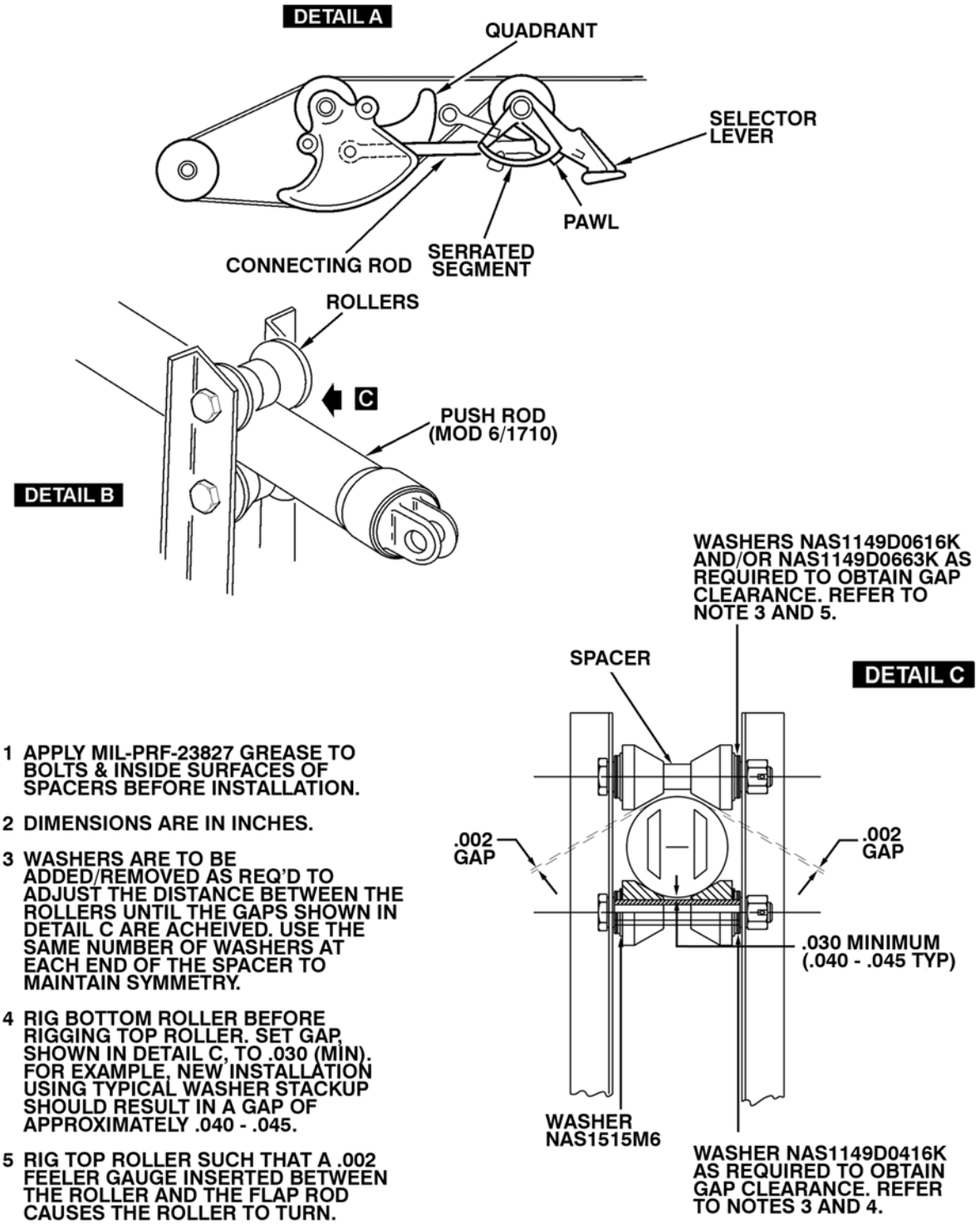
24) Install forward drip tray.

25) Install removed cabin ceiling panels.



Wing Flap System
Figure 1 (Sheet 1 of 4)

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NOTE: 1 APPLY MIL-PRF-23827 GREASE TO BOLTS & INSIDE SURFACES OF SPACERS BEFORE INSTALLATION.

2 DIMENSIONS ARE IN INCHES.

3 WASHERS ARE TO BE ADDED/REMOVED AS REQ'D TO ADJUST THE DISTANCE BETWEEN THE ROLLERS UNTIL THE GAPS SHOWN IN DETAIL C ARE ACHEIVED. USE THE SAME NUMBER OF WASHERS AT EACH END OF THE SPACER TO MAINTAIN SYMMETRY.

4 RIG BOTTOM ROLLER BEFORE RIGGING TOP ROLLER. SET GAP, SHOWN IN DETAIL C, TO .030 (MIN). FOR EXAMPLE, NEW INSTALLATION USING TYPICAL WASHER STACKUP SHOULD RESULT IN A GAP OF APPROXIMATELY .040 - .045.

5 RIG TOP ROLLER SUCH THAT A .002 FEELER GAUGE INSERTED BETWEEN THE ROLLER AND THE FLAP ROD CAUSES THE ROLLER TO TURN.

6 BOLTS SHOULD NOT BE OVER TIGHTENED. ROLLERS/SPACERS SHOULD TURN FREELY ON BOLTS.

PRE MOD 6/2186

Wing Flap System
Figure 1 (Sheet 2 of 4)

NOTE: 1 APPLY MIL-PRF-23827 GREASE TO BOLTS & INSIDE SURFACES OF SPACERS BEFORE INSTALLATION.

2 DIMENSIONS ARE IN INCHES.

3 WITH ROD PUSHED ALL THE WAY INSIDE THE WING (I.E. FLAP FULLY RETRACTED). PRE-FIT ROLLERS, SPACERS, & BOLTS (NOTE: NO WASHERS) BY CENTERING THE ROD ON ROOT RIB CUT-OUT AND BETWEEN ROLLERS/SPACERS. SPACERS SHALL BE APPROX CENTERED BETWEEN SUPPORT ANGLES.

4 SLIDE ROLLERS TO MAKE CONTACT WITH ROD, THEN MEASURE GAPS G1, G2, G3 & G4. REMOVE PRE-FIT ROLLERS, SPACERS & BOLTS.

5 DETERMINE QUANTITY OF NAS1149D0663K AND NAS1149D0616K WASHERS USING DIMENSION: Gx-.05 = Tx; CHOOSE EQUAL OR NEXT GREATER THICKNESS.

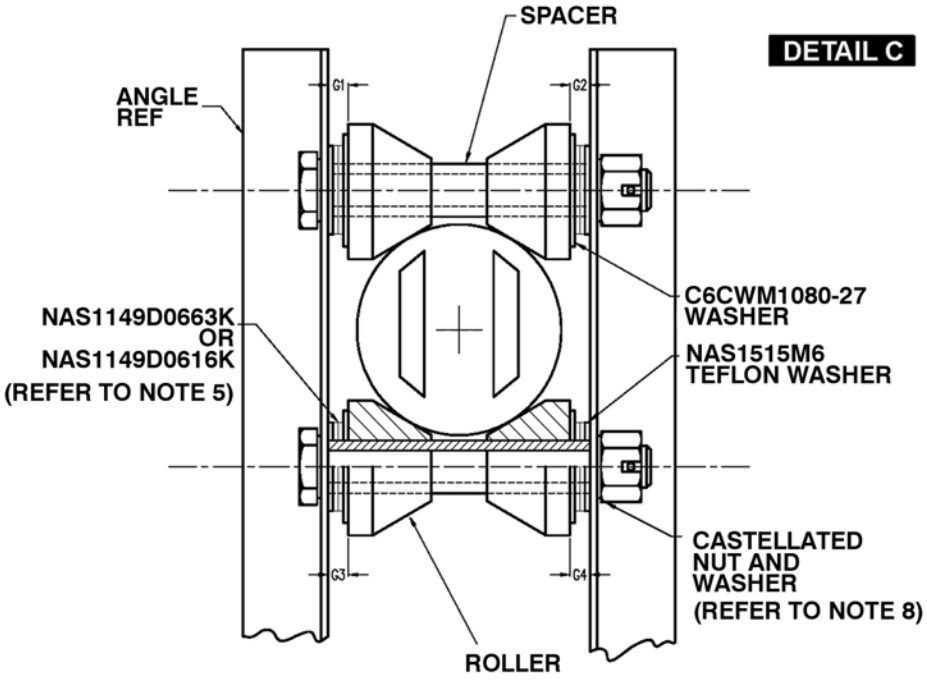
EXAMPLE:
G1=.189, THEREFORE G1-.05 = .139
REQD WASHER: 2x NAS1149D0663K = .126
1x NAS1149D0616K = .016
.142

6 INSTALL BOTTOM ROLLER/SPACER WITH WASHERS AS PER NOTE 5. ENSURE THE SPACER IS APPROX CENTERED: I.E. EQUAL GAP BETWEEN FWD & AFT SUPPORT ANGLE.

7 WHILE PUSHING THE FLAP ROD DOWNWARD TO COMPRESS THE LOWER ROLLERS, INSTALL THE UPPER ROLLERS/SPACER AS PER NOTE 5. CENTER THE SPACER ON THE FLAP ROD.

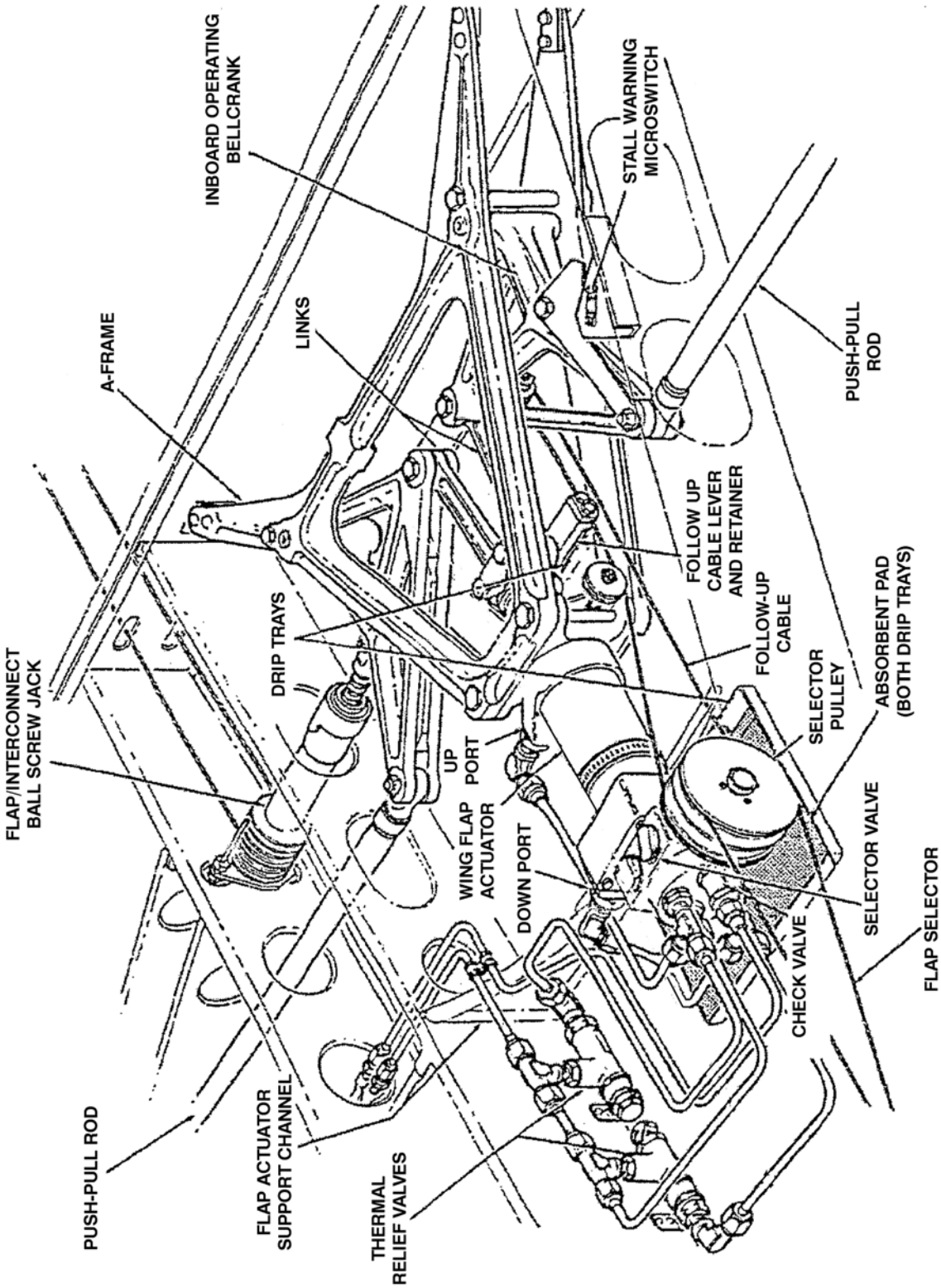
8 FIRST, TIGHTEN BOTH NUTS TO ENSURE A SNUG CONTACT AGAINST THE WASHER/BRACKET. THEN, TIGHTEN BOTH NUTS BY 1 TO 3 CASTELLATED SLOTS TO ALIGN COTTER PIN HOLE ON THE BOLT. IF MORE THAN 50% OF PIN HOLE IS ABOVE THE NUT CASTELLATION, ADD NAS1149D0463K OR NAS1149D0416 WASHER UNDER NUT.

9 MAKE SURE THE ROLLERS/SPACERS ARE TURNING FREELY ON BOLTS (ROD NOT SLIDING) WHILE THE FLAP RODS MOVE IN AND OUT.



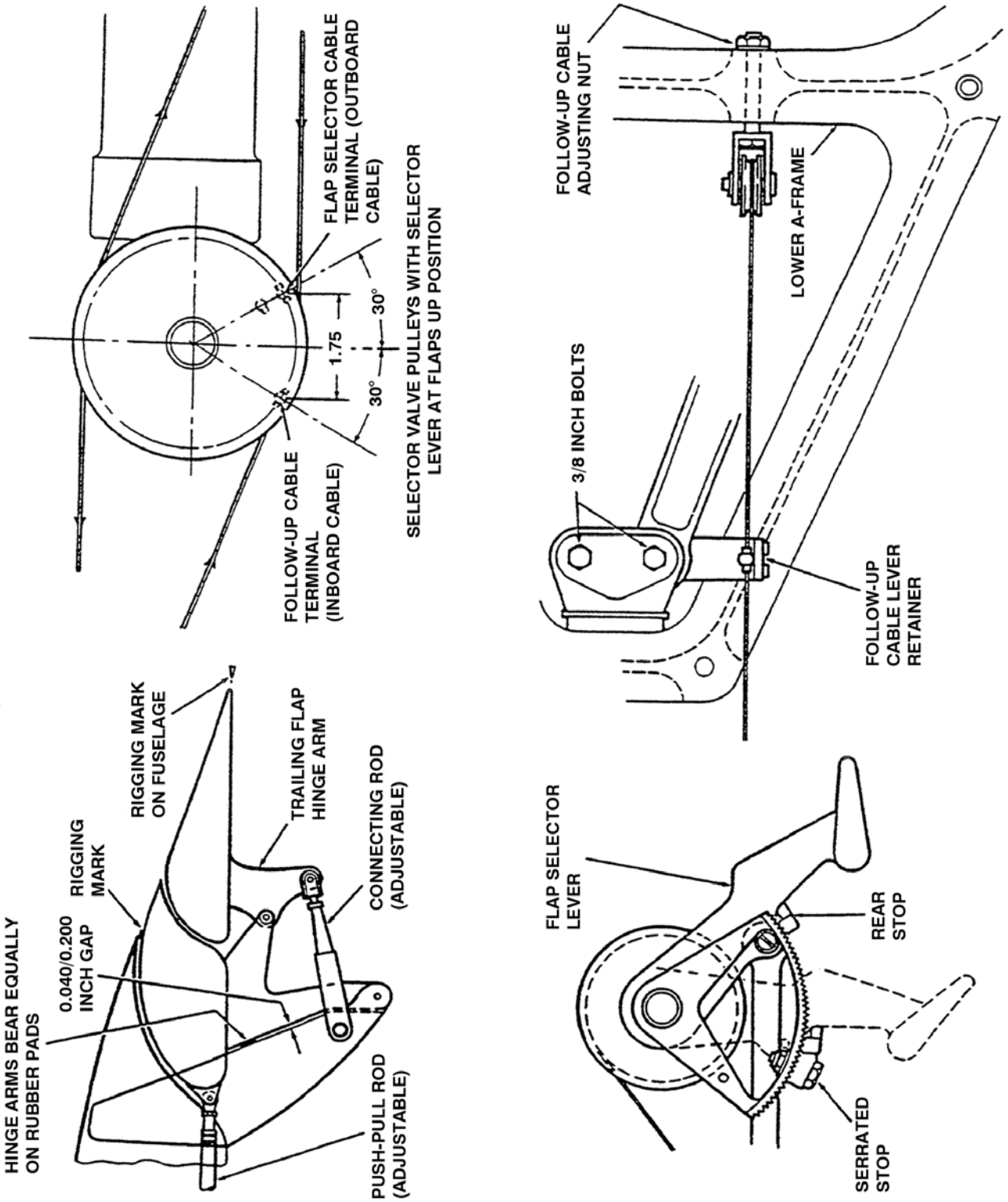
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Wing Flap System
Figure 1 (Sheet 3 of 4)



Wing Flap System
Figure 1 (Sheet 4 of 4)

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Wing Flaps System – Rigging
Figure 2

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2. Adjustment/Test

A. Bench Pressure Test

Equipment required:

- Suitable hydraulic test rig capable of supplying and maintaining 2300 psi. Clean hydraulic fluid (MIL–H–5606).

1) Compress piston rod into flap actuator; leave down port open, and connect test rig to up port. (Refer to Figure 2).

2) Apply 2300 psi hydraulic pressure for three minutes; there must be no external leakage from open down port.

3) Leave up port open, and connect test rig to down port; fully extend piston rod.

4) Apply 2300 psi hydraulic pressure for three minutes; there must be no leakage from open up port or cylinder.

5) Release hydraulic pressure and check cylinder for permanent deformation.

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